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**Proof of Tanaka's Central Limit Theorem**



# 1 Introduction

**In this report, we aim to provide a proof of Tanaka's Central Limit Theorem. The statement of the theorem is as follows:**

$$\mathrm{Law}\left(\frac{S_n - \mathbb{E}[S_n]}{\sqrt{\mathrm{Var}(S_n)}}\right) \xrightarrow{W_2} \mathcal{N}(0,1) \qquad \text{as} \quad n \longrightarrow \infty \tag{1}$$

**This theorem states that the second-order Wasserstein distance between the sum of $n$ independent, identically distributed random variables and the normal distribution tends to zero as $n$ increases. To prove this theorem, we first introduce a suitable metric for measuring the distance between two distributions, and then show that the distance between the sum of independent, identically distributed random variables and the normal distribution becomes progressively smaller.**
**In classical convergence (convergence in distribution), it is not possible to specify a particular rate of convergence. For example, one distribution may converge to the normal distribution much faster than another, but classical convergence provides no information about this distinction. In this type of convergence, there is no quantitative notion of distance; we can only say that as $n$ increases, the shape of the distribution approaches that of a normal distribution. However, if we can define a precise numerical notion of distance between two distributions, we can study convergence quantitatively and obtain a stronger result.**
**We first intend to present the basic concepts of measure theory. We then define random variables and expectation. Next, using these concepts, we introduce couplings and distances between distributions, and finally outline a general approach to proving Tanaka's Central Limit Theorem.**

# 2 Measure-Theoretic Framework

## 2.1 Sigma-Algebra:

**Given a nonempty set $S$, a collection of its subsets that contains $\emptyset$ and $S$ and is closed under countable unions, countable intersections, and complementation is called a sigma-algebra ($\mathcal{S}$).**
**The elements of a sigma-algebra are called measurable sets. The pair $(S, \mathcal{S})$ is called a measurable space.**

## 2.2 Definition of a Measure:

**On the measurable space $(S, \mathcal{S})$, a mapping $\mu : \mathcal{S} \to [0, \infty)$ is called a measure if it satisfies the following two properties:**

- $\mu(\emptyset) = 0$
- **For countably many pairwise disjoint sets $A_i$ (whose pairwise intersections are empty):** $\mu\left(\bigcup_{i\geq 1} A_i\right) = \sum_{i\geq 1} \mu(A_i)$

**If $\mu(S) = 1$, then $\mu$ is called a probability measure.**

## 2.3 Probability Space:

**The triple $(\Omega, \mathcal{S}, \mathbb{P})$ is called a probability space. Here, $\Omega$ is the sample space, $\mathcal{S}$ is the sigma-algebra, and $\mathbb{P}$ is the probability measure.**

### 2.4 Measurable Functions:

**Given two measurable spaces $(S_1, \mathcal{S}_1)$ and $(S_2, \mathcal{S}_2)$, a function $f : S_1 \to S_2$ is measurable if and only if:**

$$f^{-1}(B) \in \mathcal{S}_1 \text{ for all } B \in \mathcal{S}_2$$

### 2.5 Random Variable:

**On the probability space $(\Omega, \mathcal{A}, \mathbb{P})$, a measurable function $X : \Omega \to S$ is called a random variable, where $(S, \mathbb{S})$ is a measurable space. Furthermore:**

$$\mathbb{P}(X = x) = \mathbb{P}(X^{-1}(x)) = \mathbb{P}(\omega \in \Omega : X(\omega) = x)$$
$$\mathbb{P}(X \in B) = \mathbb{P}(X^{-1}(B)) = \mathbb{P}(\omega \in \Omega : X(\omega) \in B)$$

### 2.6 Law (Distribution) of a Random Variable:

**If $X : \Omega \to S$ is a random variable defined on the probability space $(\Omega, \mathcal{A}, \mathbb{P})$, and $(S, \mathcal{S})$ is a measurable space, then the distribution of $X$ is the function $\mu : S \to [0, 1]$ defined as follows:**

$$\mu(B) = \mathbb{P}(X^{-1}(B)) \tag{2}$$

**In fact, $\mathbb{P}$ is a mapping that assigns a probability to subsets of $\Omega$ (which are elements of the sigma-algebra $\mathcal{A}$). The function $\mu$, however, assigns probabilities to subsets of $S$ (which are elements of the sigma-algebra $\mathcal{S}$).**

### 2.7 Monotone and Dominated Convergence:

**For arbitrary measurable functions $f$ and $f_n$ from $S \to \mathbb{R}$, if $f_n$ converges monotonically to $f$, then $\int_S f_n \, d\mu$ converges monotonically to $\int_S f \, d\mu$. This is called the Monotone Convergence Theorem.**
**For arbitrary measurable functions $f$ and $f_n$ from $S \to \mathbb{R}$, if $f_n < g$ and $\int_S |g| \, d\mu < \infty$, then $\int_S f_n \, d\mu$ converges to $\int_S f \, d\mu$. This is called the Dominated Convergence Theorem. These two theorems will be used at the end to prove Tanaka's Central Limit Theorem, by showing that the sum of distributions eventually converges to a single distribution.**

### 2.8 Expectation:

**The expectation (mean) of a random variable $X$ is defined as follows:**

$$\mathbb{E}X = \int_\Omega X \, d\mathbb{P} \tag{3}$$

## 3 Probability Metrics and Wasserstein Distance

**To quantitatively examine the difference between two distributions, we must first specify the notion of distance. Different notions of distance can be considered, each measuring the difference in a particular way. In the following, we describe two methods for measuring distance:**

### 3.1 Total Variation Distance:

The Total Variation Distance between probability measures $\mu_1$ and $\mu_2$ on the measurable space $(S, \mathcal{S})$ is defined as follows:

$$d_{\mathrm{TV}}(\mu_1, \mu_2) = \sup_{A \in \mathcal{S}} |\mu_1(A) - \mu_2(A)|$$

**If $\mu_1$ and $\mu_2$ have density functions $f_1$, $f_2 : S \to \mathbb{R}_+$ with respect to a reference measure $\nu$ on the measurable space $(S, \mathcal{S})$, then:**

$$d_{\mathrm{TV}}(\mu_1, \mu_2) = \frac{1}{2} \int_S |f_1(x) - f_2(x)| \; \nu(dx).$$

**That is, the total variation distance between two distributions is half the difference between their probability masses.**

### 3.2 Coupling:

**Having introduced the basic concepts, we now turn to the concept of coupling. A coupling of the probability measures $\mu_1$ on $(S_1, \mathcal{S}_1)$ and $\mu_2$ on $(S_2, \mathcal{S}_2)$ is a pair $(X_1, X_2)$ of random variables defined on a common probability space $(\Omega, \mathcal{S}, \mathbb{P})$ such that the distribution of $X_1$ equals $\mu_1$ and the distribution of $X_2$ equals $\mu_2$. In general, a coupling is a joint probability distribution of two random variables on the product space $(S_1 \times S_2)$, whose marginal distributions are $\mu_1$ and $\mu_2$; their dependence is determined by the choice of coupling.**

**We now provide an upper bound for the total variation distance:**

**The total variation distance between the probability measures $\mu_1, \mu_2$ on the measurable space $(S, \mathcal{S})$ is bounded as follows:**

$$d_{\mathrm{TV}}(\mu_1, \mu_2) \leq \mathbb{P}(X_1 \neq X_2)$$

**for every possible coupling $X_1, X_2$ with distributions $\mu_1, \mu_2$. We also know that there always exists a coupling for which equality holds in the above relation.**

### 3.3 Wasserstein Distances:

**In the previous subsection, we discussed the total variation distance as a way to compare the difference between two distributions. In this subsection, we introduce another interpretation of distance and another way to measure their difference.**

**The Wasserstein distance of order $p \in [1, \infty)$ between the probability measures $\mathcal{V}, \mu$ on $\mathbb{R}$ is defined as follows:**

$$W_p(\mu, \mathcal{V}) = \inf_{(X,Y)\in\Gamma(\mu,\mathcal{V})} (\mathbb{E}|X - Y|^p)^{\frac{1}{p}}$$

**As we can see, both this distance and the distance defined earlier stem from the following idea of measuring distance:**

$$\inf_{(X,Y)\in\Gamma(\mu,\mathcal{V})} \mathbb{E}(cost(X, Y))$$

**That is, we calculate the infimum of the expected transportation cost over all possible couplings, using different cost functions.**

**If we choose the cost function to be $\mathbb{1}_{(X)}$, we obtain:**

$$\inf_{(X,Y)\in\Gamma(\mu,\mathcal{V})} \mathbb{E}(cost(X, Y)) = \inf_{(X,Y)\in\Gamma(\mu,\mathcal{V})} \mathbb{E}\, 1_{(X\neq Y)} = \inf_{(X,Y)\in\Gamma(\mu,\mathcal{V})} \mathbb{P}(X \neq Y)$$

**As can be seen, choosing the cost function in this way leads to the distance expressed by $d_{\mathrm{TV}}$.**

**If we choose a different cost function, such as $cost(X, Y) = |X - Y|^p$, which is called the $p$-th moment, we obtain the following definition:**

$$\inf_{(X,Y)\in\Gamma(\mu,\mathcal{V})} \mathbb{E}(cost(X, Y)) = \inf_{(X,Y)\in\Gamma(\mu,\mathcal{V})} \mathbb{E}\,|X - Y|^p$$

**As we can see, this cost function leads to the definition of the Wasserstein distance (the exponent $\frac{1}{p}$ is used to make the expression a distance function with the correct unit).**

**The Wasserstein distance of order $p$ is a metric on the space of all distributions on $(R, \beta(R))$ whose $p$-th moments are finite (bounded).**

### 3.4 Definition of the Quantile Function:

**The Quantile function is the generalized inverse of the cumulative distribution function; that is, $(0 \leq a \leq 1)$ $Q(a)$ gives us a point at which the cumulative distribution function equals $a$.**

**Using this function, we can express the Wasserstein distance as follows, for $p \in [1, \infty)$:**

$$W_p(\mu_1, \mu_2) = (E(Q_1(U) - Q_2(U))^p)^{\frac{1}{p}}$$

**In the formula above, $Q_1, Q_2$ are the Quantile functions corresponding to $\mu_1, \mu_2$, respectively, where $U$ is a uniformly distributed random variable on $(0, 1)$.**

**It can be seen that this definition states that $(Q_1, Q_2)$ is an optimal coupling for the Wasserstein distance of order $p$.**

# 4 Proof of Tanaka's Central Limit Theorem

We first restate the theorem. If $S_n = X_1 + ... + X_n$ is the sum of independent and identically distributed $(i.i.d)$ random variables and $\mathbb{E}X^2 < \infty$, then:

$$\text{Law}\left(\frac{S_n - \mathbb{E}[S_n]}{\sqrt{\text{Var}(S_n)}}\right) \xrightarrow{W_2} \mathcal{N}(0,1) \qquad \text{as} \quad n \longrightarrow \infty$$

**This theorem states that as $n$ increases, the Wasserstein distance between the distribution of $S_n$ (after normalizing its mean to $0$ and its variance to $1$) and the normal distribution $\mathcal{N}(0,1)$ tends to zero. To prove this theorem, we follow these steps:**

- First, we find an optimal coupling and use it to calculate the Wasserstein distance between the distribution of interest and the normal distribution.
- Next, we use a very important inequality stating that the sum of the Wasserstein distances of two distributions from the normal distribution is always greater than the Wasserstein distance between their sum and the normal distribution, unless the distributions themselves are normal.
- In the next step, we show that repeatedly summing these random variables eventually leads to a limiting distribution.
- Finally, we show that after these distributions are summed, their Wasserstein distance from the normal distribution decreases at each stage and ultimately tends to zero.

**For convenience, we will use the notation $\epsilon[.]$ in the proof. This symbol represents the second-order Wasserstein distance between an arbitrary distribution and the corresponding normal distribution with the same variance.**

**In the previous section, we saw that the optimal coupling is defined as follows:**

$$W_2^2(X,Y) = \mathbb{E}(Q_1(U) - Q_2(U))^2$$

**where $U \sim Uniform(0,1)$ and $Q1$ and $Q_2$ are the quantile functions of $\mu_1$ and $\mu_2$. Consequently, the optimal coupling used in this proof is $\epsilon[f] = \mathbb{E}(|X-Y|^2)$ if and only if $X = F^{-1}(G(Y))$, where $F^{-1}$ is the quantile function of the distribution function $F$ and $G$ is the Gaussian distribution with the same variance as $F$. We now introduce the following inequality, which will be used in the proof. For any nonzero real numbers $a$ and $b$ and independent random variables $X$ and $Y$ such that $\mathbb{E}[X] = \mathbb{E}[Y] = 0$, we have:**

$$\epsilon[aX + bY] < a^2\epsilon[x] + b^2\epsilon[Y]$$

**unless both are Gaussian, in which case the inequality becomes an equality. This inequality states that if we combine two distributions, the resulting distribution becomes closer to the Gaussian distribution. For example, consider the two distributions $\delta_1$ and $\delta_{-1}$. Clearly, the combination of these two distributions $(\frac{1}{2}\delta_1 + \frac{1}{2}\delta_{-1})$ is closer to the Gaussian distribution than either of these distributions individually.**

**We now use the Monotone and Dominated Convergence Theorems to introduce a limit theorem for the proof. If $f_n$ approaches a function $f \in \mathcal{P}$ from above such that:**

$$\lim_{N\to\infty} \sup_{n\geq 1} \int_{|x|>N} x^2 \, f_n(dx) = 0$$

**then $\lim_{n\to\infty} \epsilon[f_n] = \epsilon[f]$. This theorem holds when the $p$-th moments of the distributions $f_n$ are uniformly bounded in $n$ for some $p > 2$. In this proof, we will use the boundedness of the fourth moment to apply this result. In fact, this theorem states that to find the limit of $\epsilon[f_n]$, it is sufficient to calculate $\epsilon[f]$, the Wasserstein distance between the limiting function and the Gaussian function. In other words, this theorem allows us to interchange the limit and $\epsilon[.]$.**

**We now begin the proof of Tanaka's Central Limit Theorem:**

**Suppose we have independent and identically distributed random variables $X_1, ..., X_2$ with mean $0$ and variance $1$. Tanaka's Central Limit Theorem states that the distribution of $\xi_n = \frac{X_1+...+X_n}{\sqrt{n}}$ converges to the normal distribution as $n$ increases ($n \to \infty$).**

**In this proof, we show that $\epsilon[\xi] \to 0$ as $n \to \infty$. We first establish the boundedness of the fourth moment.**

**Assume that $\mathbb{E}(X_1^4) < \infty$. It can easily be shown that $\mathbb{E}(\xi_n^4) = \frac{1}{n}(X_1^4) + 3\left(1 - \frac{1}{n}\right)$. Therefore, $\mathbb{E}(\xi_n^4) \leq C$, giving an upper bound for the fourth moment. We now prove the result only for $n = 2^k$ and then extend it to all $n$. We define the new sequence $\eta_k = \xi_{2^k}$.**

**By Tanaka's inequality, we have:**

$$\epsilon[aX + bY] < a^2\epsilon[X] + b^2\epsilon[Y]$$

**We generate an independent copy of $\xi_{2^k}$ and denote it by $\xi'_{2^k}$.**

By the convergence theorems, it is clear that as $k \to \infty$, $\epsilon(\eta_k)$ converges monotonically to a number such as $\ell$. If we denote the limiting distribution of $\eta_k$ as $k \to \infty$ by $\mu$, and consequently consider independent random variables $\eta$ and $\zeta$ with distribution $\mu$, we have (because of the convergence of the fourth moment, we can move the limit inside $\epsilon$):

$$\epsilon(\eta) = \lim_{k\to\infty} \epsilon(\eta_k) = \ell$$

Also, since $\eta$ is the limit of the sequence $\{\eta_k\}$ at infinity, we have: $\ell$ is a constant.

$$\epsilon\left(\frac{\eta + \zeta}{\sqrt{2}}\right) = \ell$$

Thus, we have

$$\epsilon\left(\frac{\xi + \zeta}{\sqrt{2}}\right) = \epsilon(\eta)$$

Therefore, $\eta$ must be normally distributed, that is,

$$\ell = 0$$

**It remains only to extend the result to all $n$. We write any $n \geq 1$ as follows:**

$$n = \sum_{k=0}^{m} \varepsilon_k \, 2^k, \qquad \varepsilon_k \in \{0, 1\}$$

**In fact, we express an arbitrary $n$ as a sum of powers of two. Then, using the preceding inequality for this sum, we obtain:**

$$\epsilon[\xi_n] \leq \frac{1}{n} \sum_{k=0}^{m} \varepsilon_k \, 2^k \epsilon[\xi_k]$$

**We have seen that for large $k$, $\epsilon[\xi_k]$ tends to zero, and for terms with small $k$, their coefficient (weight) $\left(\frac{1}{n} \sum_{k=0}^{m} \varepsilon_k \, 2^k\right)$ also tends to zero as $n$ increases. Finally, we conclude that:**

$$\epsilon[\xi_n] \longrightarrow 0 \quad \textbf{as} \quad n \longrightarrow \infty$$

**Thus, the theorem is proved.**

# 5 References


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